

Hi! I am a PhD candidate specializing in Systems and Data Center Networking. In my current work, I focus on developing frameworks to make congestion control evaluations more representative, and hence better guide the design of high-performance network algorithms.

EDUCATION

Carnegie Mellon University

PhD in Electrical and Computer Engineering

Advisor : Prof. Theophilus Benson

CGPA : **4.00**/4.00

Pittsburgh, PA, USA

2024-2029

Indian Institute of Technology, Hyderabad

Bachelor of Technology in Electrical Engineering, with minor in Artificial Intelligence

CGPA : **9.75**/10.00

President of India Gold Medallist

Hyderabad, India

2020 - 2024

City International School, Pune

High School and Senior Secondary Education (Science Discipline)

Grade 12 : **95%** Grade 10 : **97%**

Pune, India

2007-2020

RESEARCH AREAS AND PUBLICATIONS

Data Center Congestion Control

Projects

Evaluating Congestion Control Under Incast Bursts

Oct 2025 – Present

with Prof. Theo Benson and Srikanth Sundaresan (Meta)

- A CCA-aware evaluation framework that maps incast burst parameters to key burst dimensions, enabling systematic coverage of congestion control behaviors under incast workloads.

Publications

- *Towards Truly Burst Aware Evaluation of Data Center Congestion Control (2026)*
Asia-Pacific Workshop on Networking (APNet)

eBPF Management Abstractions

Projects

Prism: Towards Software Defined Observability

Feb 2025 – Oct 2025

with Prof. Theophilus Benson at CMU

- Introduces a reactive programming model for observability using eBPF probes by decoupling probe specification from execution.

Apiary: A Runtime Framework for Distributed eBPF Program Management

Aug 2024 – Oct 2025

with Prof. Theophilus Benson at CMU

- Proposes abstractions for distributed eBPF applications, addressing isolation and lifecycle management in large-scale deployments.

Leveraging Bytecode Understanding for Debugging eBPF NF Deployments

Sep 2023 – May 2024

with Dr. Praveen Tamma and Dr. Ramakrishna Upadrasta at IIT Hyderabad

- Built a system that analyzes packet behavior using only eBPF bytecode, enabling debugging without source code access.

eBPF Program Registry and Code Summarization

Apr 2023 – Dec 2023

with Dr. Palanivel Kodeswaran and Dr. Sayandeep Sen (IBM Research Lab)

- Developed an open-source registry with LLM-based summarization and Elasticsearch-powered semantic search for eBPF programs.

Publications

- *Yaksha-Prashna: Understanding eBPF Bytecode Network Function Behavior* **(2026)**
arXiv Preprint
- *Apiary: Distributed Programming and Lifecycle Management for eBPF* **(2025)**
N2Women Workshop @ SIGCOMM 2025

Pulsar Astronomy

Publications

- *The Indian Pulsar Timing Array Data Release 2: I. Dataset and Timing Analysis* **(2025)**
Publications of the Astronomical Society of Australia
- *Low-frequency pulse-jitter measurement with the uGMRT I: PSR J0437–4715* **(2024)**
Publications of the Astronomical Society of Australia
- *Application of Efron-Petrosian method to radio pulsar fluxes* **(2024)**
Journal of Cosmology and Astroparticle Physics
- *Do Pulsar and Fast Radio Burst dispersion measures obey Benford's law?* **(2023)**
Astroparticle Physics

HONORS AND AWARDS

Prabhu and Poonam Goel Graduate Fellowship

Mar 2026

Awarded by the CMU ECE Department.

Carnegie Institute of Technology Dean's Fellow

Aug 2024 - July 2025

Awarded full tuition support and stipend for one year of PhD study.

President of India Gold Medal

July 2024

For achieving the highest overall CGPA among all graduating B.Tech students at IIT Hyderabad.

Institute Silver Medal

July 2024

For achieving the highest CGPA in the Electrical Engineering Department at IIT Hyderabad.

Academic Excellence Award

Apr 2024, Apr 2022

For achieving the highest CGPA in Electrical Engineering Department at IIT Hyderabad, in the calendar years 2023 and 2021.

Excellence in Innovation Award

Jan 2022

Awarded for being one of top 5 among 1,300 participants in the Road2Shine program, a joint venture by the Indian and Japanese governments.

All India Rank of 3418 in JEE-Advanced

Nov 2020

among 1 Million Candidates taking the test.

Merit Certificate by CBSE

Apr 2018

For scoring in the top 0.1% among 1.7 Million candidates in Science in Grade 10.

PROFESSIONAL SERVICE AND OUTREACH ACTIVITIES

Artifact Evaluation Committee Member, CAIS

2026

Served as part of Artifact Evaluation Committee for the first ACM Conference on AI and Agentic Systems.

K-12 Community Outreach

2026

Goals : Making engineering more accessible and equitable in K-12 spaces, re-envisioning engineering education pedagogy.

Shadow PC Member, IMC

2026

Shadow PC Member, ACM CoNEXT

2025

Evaluator, Meeting of the Minds – Carnegie Mellon University	2025
Reviewed senior undergraduate research projects as part of CMU’s annual research symposium.	
Yoga Instructor at CMU	2024 - Present
Teach weekly group yoga sessions as part of CMU’s GroupX fitness program.	
Volunteer Teacher - National Service Scheme, India	2022-2024
Volunteered as a teacher and motivational lectures lead of Aksharamaala Program in India, teaching Grade 10 Mathematics at rural schools.	

WORK EXPERIENCE

Software Developer	May 2023 – July 2023
<i>Arcesium</i>	<i>Hyderabad, India</i>
Developed an application to analyze the CI/CD workflow in a Dev-Ops setting. This involved extensive work with APIs and dashboards like Apache Superset and Grafana.	
Undergraduate Researcher	Dec 2022
<i>University of Tokyo</i>	<i>Tokyo, Japan</i>
Visited UTokyo for a research internship under the guidance of Prof. Kunihiro Sadakane. The project was based on succinctly encoding ordinal trees using tree covering.	
Student Member	Jun 2022 - Jun 2024
<i>Indian Pulsar Timing Array</i>	<i>Hyderabad, India</i>
I was a student member of the InPTA Consortium, as part of the Data Reduction team. I also worked with National Centre for Radio Astrophysics for collating legacy data of the uGMRT (Giant Metrewave Radio Telescope).	
5G Testbed Intern	Jan 2022 – Jun 2022
<i>Wisig Networks</i>	<i>Hyderabad, India</i>
Developed a module to transmit DCI (Downlink Control Information) from gNB Base station to User equipment. Gained knowledge about L2/L3 protocol stack development.	